

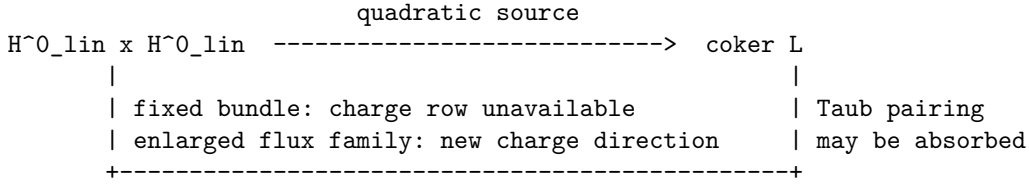
Charge fibres and Taub obstructions in compact Einstein–Maxwell gravity

Bridge note for mathematical relativity and linearization-stability researchers. Status: draft allowed, 17 July 2026.

1. Their object and the unresolved question

Linearization stability asks whether every solution of the linearized field equations is tangent to an actual family of exact solutions. On compact Cauchy surfaces with symmetries, quadratic Taub constraints can obstruct such an extension. The modern genericity result of Saraykar and Rai places this in the class of constant-mean-curvature spacetimes with compact Cauchy hypersurfaces ([arXiv:1609.07703](#)); it builds on the Fischer–Marsden–Moncrief tradition.

Our adjacent question is whether the obstruction depends on the **global charge fibre**: is a tangent obstructed on a fixed $U(1)$ bundle but extendible when electric or magnetic flux is admitted as part of the family?



2. Exact dictionary

Item	Linearization-stability language	This project	Conversion or caveat
Background	Einstein or Einstein–matter solution	compact $\mathbb{R} \times S^1 \times S^2$ Einstein–Maxwell product	It is simultaneously a tuned Weyl–Maxwell background.
Linear tangent	kernel of the linearized equations modulo gauge	standard photon/graviton harmonics and extra Weyl classes	The actual product-background stabilizers are retained as global symmetries; no absolute quotient is authorized.
Quadratic constraint	Taub charge paired with a Killing initial datum	constant-lapse component of $D^2 E[\Phi_1, \Phi_1]$	Other adjoint-cokernel classes remain to be classified.
Global sector	fixed conserved charges	fixed Chern class P_N , with electric variation tracked separately	A continuous magnetic variation changes the bundle family.
Extension	solution of $L\Phi_2 = -\frac{1}{2}D^2 E[\Phi_1, \Phi_1]$	explicit correction or normalized adjoint witness	A removable output block is not proof that the real tangent extends.
Pairing	constraint/cokernel pairing	relative Taub form $\mathfrak{D}$	Exact algebraic arithmetic is used.

3. Reproduced benchmark

The compact topology fixes the magnetic sector. On the rational fixture the bundle number expands as

$$N(\epsilon) = 2 + 2\epsilon^2 p.$$

A smooth family on the same $N = 2$ bundle therefore has $p = 0$. The constant-lapse Taub component cannot be removed by silently varying magnetic flux. On the declared four-dimensional fixture span—radion, duality, $\ell = 1$ photon, and plus-branch $\ell = 2$ graviton—the symmetric form is

$$\text{diag}\left(-2, -\frac{1}{2}, -\frac{16}{3}, -\frac{72}{5} - 12\sqrt{3}\right).$$

Gauge descent and Cauchy-slice conservation follow from the coupled action Noether identities. The benchmark is independently replayed by:

```
python3 -m bridge.einstein_sector.compact_harmonic_domain_taub_descent \
--verify bridge/certificates/compact_harmonic_domain_taub_descent.json
python3 bridge/einstein_sector/verify_compact_harmonic_domain_taub_descent.py
```

4. Added result

Scoped charge-fibre obstruction. On the declared compact fixed bundle, several physical photon, graviton, and extra-Weyl tangents have nonzero second-order Taub pairings and therefore no smooth periodic second-order correction. Allowing a formal magnetic coefficient can remove specified constant-lapse components, but that coefficient belongs to an enlarged flux family, not the original fixed-bundle phase space.

The most recent real test uses the degenerate axial-polar $\ell = 2, k = 0$ minus-frequency Einstein pair. Its Hermitian Taub form is positive definite, so every nonzero real combination is obstructed at fixed bundle topology. A separate axial-polar sum-frequency source has an exact correction; it does not remove the zero-frequency constraint arising from conjugate self-products. Thus source solvability must be checked in every selection channel before declaring extension.

There is now also an all-generic extra-sector theorem. On the closed slice, the covariant Hamiltonian identity gives

$$\left\langle \zeta_X, \frac{1}{2} D^2 E_{\text{WM}}[u, u] \right\rangle = \mu_X(u) = \frac{1}{2} \Omega_{\text{WM}}(u, \mathcal{L}_X u).$$

Its normalization is fixed by exact agreement with three direct tensor Taub calculations: the axial extra block and the axial and polar Einstein-minus fixtures at $\ell = 2, k = 0$. In the real-mode convention $\Phi = \text{Re}(ce^{-i\omega t})$,

$$\mu_H = -\frac{LN_{\ell m}}{4} \omega^2 c^\dagger G c.$$

Both axial and polar extra-Weyl Gram matrices are positive definite for every $\ell \geq 2$ and every compact momentum, and $\omega_e^2 = k^2 + \ell(\ell + 1) - 2/3 > 0$. Hence the fixed-bundle Taub form is definite on the complete real pure-extra generic sector. No nonzero tangent in that sector extends to second order. This includes finite and rapidly decreasing finite-energy harmonic superpositions.

The obstruction can nevertheless be cancelled by an Einstein component. A single fixed nonzero-momentum travelling block cannot satisfy both the energy and momentum Taub constraints nontrivially, because $\omega_- < \omega_e < \omega_+$. At zero momentum, however, the axial $\ell = 2, m = 0$ Einstein-minus mode and the second axial extra mode have a nonzero common-zero combination with

$$|a_e|^2 = \frac{27}{52} (5\sqrt{3} - 6) |a_-|^2.$$

All five background-stabilizer moment maps vanish on this real tangent. Its complete quadratic source has only polar $\ell = 0, 2, 4$ outputs. Direct tensor expansion shows that the homogeneous zero-frequency sources cancel exactly; the remaining homogeneous channels have explicit corrections, while every $\ell = 2, 4$ channel is removed by an exact off-shell polar inverse. Thus this declared mixed tangent has a complete second-order correction. This is a single formal second-order extension, not an all-orders family or general closure theorem.

5. Consequence in their language

The result promotes “hold the charge fixed” from an informal side condition to part of the domain of the obstruction map. It suggests that linearization stability for gauge-gravity systems should be indexed by a charge fibre—or a stratum of the global bundle moduli—before genericity is discussed. The same linear tangent can have different integrability status in different enlarged families without contradiction.

6. Scope boundary

This is LOCAL-ALGEBRAIC/REDUCED-MODE, not a causal nonlinear evolution theorem. The pure-extra generic block is now complete, but mixed Einstein-extra cancellation is classified only for one extended fixture and one same-momentum no-go. The full common H, P_x, J_i zero locus, exceptional/Jordan modes, charge-relaxed corrections, all-orders integrability, and any justified stabilizer quotient remain open. No asymptotic, scattering, or quantum claim follows. The result does not say that the entire Einstein sector fails nonlinearly.

7. One useful question for adjacent experts

Should the fixed-versus-variable flux distinction be formulated as linearization stability on separate connected components of configuration space, as a stratified moment-map problem, or as an enlarged constraint operator with additional global rows? Which formulation best preserves the classical Taub theorem when the underlying principal bundle changes?

Reproducibility receipt

source paper: arXiv:1609.07703v1 and its cited classical framework
project certificates: COMPACT_HARMONIC_DOMAIN_AND_TAUB_DESCENT;
EINSTEIN_MAXWELL_OBSTRUCTION_BILINEAR_G1;
EINSTEIN_MAXWELL_WEYL_HERMITIAN_AXIAL_POLAR_ELL2_TAUB;
EINSTEIN_MAXWELL_WEYL_MOMENT_MAP_TAUB_BRIDGE;
EINSTEIN_MAXWELL_WEYL_MIXED_MOMENT_MAP_ZERO_LOCUS;
EINSTEIN_MAXWELL_WEYL_BALANCED_ELL0_SECOND_ORDER
verification: section 3 plus certificate-local commands
dependency tags: LOCAL-ALGEBRAIC; REDUCED-MODE
generality level: G2_GENERIC_AXIAL_POLAR_ALL_PHYSICAL_ELL_K_MOMENT_MAP_TAUB
lifecycle state: DRAFT_ALLOWED
claim flag: PURE_EXTRA_NO_GO_AND_ONE_BALANCED_MIXED_SECOND_ORDER_EXTENSION_CERTIFIED
open fields: full mixed zero locus; all-orders integrability; exceptional/global blocks